IU2541230002 - CPP01 - SUBMISSION TEMPLATE

Project Name: Library Management System
Enrollment No.: IU2541230002
Student Name: _____
Date of Submission: 13-10-2026
GitHub Repository: [Paste your GitHub repository link here, e.g. https://github.com/your-username/IU2541230002-Library-Management-System]

1. Project Summary

Problem Statement: Manual library record keeping is slow and error-prone: records get duplicated or lost, it is hard to know who holds which book, a book can be given to two people, and fines are calculated by hand. A C++ program is needed to store book, member and transaction records, issue and return books under clear rules, calculate fines, validate input and save data permanently in files.

Objectives:

- Design the system with classes, encapsulation, constructors and destructors.
- Demonstrate inheritance (Member from Person) and runtime polymorphism (virtual display()).
- Use dynamic memory (new[] / delete[]) for the record arrays.
- Implement issue and return of books with automatic fine calculation.
- Store data permanently with file handling and protect the program with validation and exception handling.

Key Features:

- Menu-driven program with 9 options: add, display, search, update, delete, issue/return, report, save/load, exit.
- Books and members can be searched by ID or keyword; delete is blocked while a book is issued or a member holds books.
- Limit of 3 books per member, 14 free days, fine of Rs. 5 per late day.
- Report with availability, fines collected and outstanding, books by category and currently issued books.
- Three data files loaded automatically at start-up; validated input (phone, e-mail, real dates).

2. Class Summary

Class	Purpose / Responsibility	Key Data Members	Key Member Functions
Person	Base class with the common details of a person; defines virtual display().	id, name, phone	Person(), Person(...), virtual display(), setName(), setPhone()
Member	Library member derived from Person; limits borrowing to 3 books.	email, borrowedCount	Member(), display(), borrowBook(), returnBook(), canBorrow()
Book	One book and its availability status.	bookId, title, author, category, available	Book(), display(), issue(), returnBook(), isAvailable()
Transaction	One issue/return record; calculates the fine.	transactionId, bookId, memberId, issueDate, returnDate	Transaction(), display(), closeTransaction(), daysLate(), calculateFine()
Library	System class: owns the dynamic arrays and applies all rules.	Book* books, Member* members, Transaction* transactions, counts, capacities	addRecord(), searchBook(), issueBook(), returnBook(), report(), saveAll(), loadAll()
FileManager	All file input/output for the three data files.	booksFile, membersFile, transactionsFile	saveBooks(), loadBooks(), saveMembers(), loadMembers(), saveTransactions(), loadTransactions()
Utils / LibraryException	Validated input, date helpers and custom exceptions.	(helper functions / exception classes)	readInt(), readPhone(), readEmail(), readDate(), LibraryException

3. Project Flow

Actual flow of the project from user input to final file storage:

USER -> MAIN MENU (main.cpp) -> LIBRARY (system class) -> BOOK / MEMBER / TRANSACTION objects -> FILEMANAGER -> FILE (books.txt, members.txt, transactions.txt)

Example (issue a book): the user chooses option 6, enters Member ID, Book ID and date; Library checks the rules, updates Book, Member and Transaction objects; saveAll() calls FileManager, which writes the three files.

4. UML Deliverables

Diagram	File Name	Completed
Use Case Diagram	UML_Diagrams/Use_Case_Diagram.png	[X]
Class Diagram	UML_Diagrams/Class_Diagram.png	[X]
Sequence Diagram 1	UML_Diagrams/Sequence_Diagram_1_Issue_Book.png	[X]
Sequence Diagram 2	UML_Diagrams/Sequence_Diagram_2_Return_Book.png	[X]

5. C++ Concepts Demonstrated

Concept	Where Used in My Project	Explanation
Encapsulation	Data members of every class	All data members are private or protected. They are read through getters and changed only through setters such as setPhone(), which reject invalid values.
Constructor	Book(), Book(...), Member(...), Transaction(...), Library()	Each entity has a default constructor (needed by new[]) and a parameterized constructor that validates its arguments and throws on bad data.
Destructor	Library::~Library(), Person::~Person()	Library's destructor releases the three dynamic arrays with delete[]. Person's destructor is virtual so deletion through a Person pointer is safe.
Inheritance	class Member : public Person	Member reuses id, name and phone from Person and adds email and borrowedCount.
Runtime Polymorphism	Person::display() (virtual), Member::display(), Library::displayRecords(), printPerson()	A Person* (or Person&) that refers to a Member calls Member::display() at run time. The base printCore() is reused by the override.
Dynamic Memory	Library constructor and growArray(); FileManager load functions	Arrays such as new Book[capacity] are created at run time because the number of records is unknown. When full, the capacity is doubled (new[], copy, delete[]). Loading from files also allocates a new array.
File Handling	FileManager (ifstream / ofstream)	Books, members and transactions are stored in three text files, loaded automatically at start-up and saved from the menu or on exit.

Concept	Where Used in My Project	Explanation
Exception Handling	LibraryException thrown in all classes; try/catch in main() and FileManager	Duplicate IDs, invalid data, rule violations and file errors throw exceptions that main() catches and reports without crashing.
Validation	Utils::readInt(), readPhone(), readEmail(), readDate()	Input is checked (numbers only, 10-digit phone, e-mail format, real calendar dates) and the user is asked again until valid.

6. Testing

Test ID	Operation	Input	Expected Output	Actual Output	Pass/Fail
T01	Add Record - Book	Book ID 101, Title "The C++ Programming Language", Author "Bjarne Stroustrup", Category "Programming"	Book is stored and a success message is shown	Book added successfully.	Pass
T02	Add Record - Member	Member ID 1001, Name "Riya Patel", Phone 9876543210, Email riya.patel@gmail.com	Member is stored and a success message is shown	Member added successfully.	Pass
T03	Duplicate ID check	Add Book with ID 101 again	Rejected with an error message	Error: A book with ID 101 already exists.	Pass
T04	Phone and e-mail validation	Phone "12345", then "98765abcde"; E-mail "riya"	Invalid values rejected and re-asked; valid values accepted	Invalid phone: enter exactly 10 digits. / Invalid email: use a format like name@example.com. Then accepted	Pass
T05	Display Records - Books	Menu 2, option 1	All books listed in a table with status	6 books shown, all Available; Total books: 6	Pass
T06	Search Record - keyword	Menu 3, keyword "programming"	Books whose title, author or category match	3 books listed (101, 102, 105); 3 book(s) found.	Pass
T07	Update Record - Book	Book ID 103; Enter, Enter; new category "Autobiography"	Only the category changes	Book updated successfully. (category later shown as Autobiography)	Pass
T08	Issue Book	Member 1001, Book 101, issue date 01-10-2026	Transaction created, book marked Issued	Book issued successfully. Transaction ID: 1	Pass
T09	Issue an already issued book	Member 1002, Book 101	Rejected, nothing changes	Error: Book is already issued to another member.	Pass

Test ID	Operation	Input	Expected Output	Actual Output	Pass/Fail
T10	Borrow limit	Member 1001 already holds 3 books; tries Book 104	Rejected (limit is 3 books)	Error: Member has reached the borrow limit of 3 books.	Pass
T11	Return Book with late fine	Book 101, return date 25-10-2026 (issued 01-10-2026)	Days late = 10, fine = 10 x Rs. 5 = Rs. 50	Days late: 10; Fine: Rs. 50.00	Pass
T12	Delete an issued book	Menu 5, Book ID 102 (currently issued)	Deletion refused	Error: This book is currently issued and cannot be deleted.	Pass
T13	Report	Menu 7 (after the above operations)	Totals, fines and issued-book list shown	Total books 5 (issued 3); fine collected Rs. 50.00; fine outstanding Rs. 45.00	Pass
T14	Save / Load persistence	Add Book 107 without saving, then Menu 8, option 2 (Load)	Data reloaded from files; unsaved Book 107 is gone	Data loaded: 5 book(s), 2 member(s), 5 transaction(s). Book 107 not listed	Pass
T15	Invalid menu input	"abc", "0", "10" at the main menu	Each rejected and the menu choice asked again	Invalid input: enter a whole number between 1 and 9. (three times)	Pass

7. Deployment

Compiler / IDE Used: g++ (GCC 13.3, C++11 standard); the code also compiles with MinGW g++, Code::Blocks, Dev-C++ and Visual Studio Code.
Compilation Command: g++ -std=c++11 -o LibraryManagement main.cpp Library.cpp FileManager.cpp Book.cpp Member.cpp Person.cpp Transaction.cpp Utils.cpp (on Windows add .exe to the output name, or run build.bat).
Executable Name: LibraryManagement.exe on Windows (LibraryManagement on Linux).
Data File(s): books.txt, members.txt, transactions.txt (kept in the same folder as the program).

8. Student Declaration

I understand the submitted project and can explain its classes, data members, member functions, file handling and UML diagrams.

Student Signature: _____